Thymulin Available for Research Use Only

(Acute infections: 0.5 mg daily up to 14 days; Healthspan: 0.25 mg daily)

Thymulin is a endogenous zinc-dependent thymic regulatory hormone that modulates T-cell function and inflammatory signaling. It is not a receptor agonist. Its zinc-bound form (Zn-thymulin) regulates thymic and peripheral immune function. Its aging rationale is supported by endogenous hormone decline and direct experiments in aged human cells and mice.

💉 💉 Probably safe

⭐ ⭐ Likely effective for healthspan and aging support

⭐ Plausibly effective for acute infections

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

This document was updated 7-Sep-26. Research survey; proposed dosing patterns are experimental extrapolations.

Human safety data are lacking for long-term use. Interactions with immunosuppressants and biologics are not well characterized.

Human research links reduced endogenous active thymulin with aging, malnutrition, and zinc deficiency. The July 2026 aging experiments strengthen the rationale for immune regulation and healthspan. (Prasad et al., 1988; Boukaïba et al., 1993; Kanemaru et al., 2026.)

Zn-thymulin vs Zinc-free FTS

In this review, zinc-free FTS denotes the uncomplexed peptide; a product labeled “thymulin” should not be assumed to contain preformed Zn-thymulin. (Bach and Dardenne, 1989; Dardenne et al., 1982.)

Thymulin biological activity requires one bound zinc ion per peptide molecule. In old-mouse spleen cells, Zn-thymulin restored impaired NK-cell activity in vitro, whereas zinc-free FTS and seperately zinc alone did not. (Dardenne et al., 1982; Muzzioli et al., 1992.)

Aging, Immunosenescence, and Inflammaging

Active thymulin declines with age, with contributions from thymic involution, reduced peptide production, and altered zinc availability or saturation. Adding zinc to serum or restoring deficient zinc status can recover part of the activity. Thus, replacing an age-depleted regulatory signal is a

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particularly direct aging hypothesis. (Prasad et al., 1988; Boukaïba et al., 1993; Mocchegiani and Fabris, 1995.)

Kanemaru et al., published July 21, 2026 in Nature Communications, found an age-associated inflammatory myeloid phenotype in humans and mice. Thymulin treatment of human peripheral blood mononuclear cells (PBMCs) ex vivo for 24 hours reduced inflammatory cytokine expression in an age-dependent manner, including samples from adults aged 52–86.

In aged mice, thymulin suppressed inflammatory myeloid-cell activity through canonical NF-κB inhibition and enhanced T-cell IFN-γ responses and cytotoxic function. In aged tumor-bearing mice it improved tumor control and response to anti-PD-L1 treatment, with much smaller effects in young animals. The short immune experiments used daily dosing for one week; tumor experiments extended daily exposure to four weeks. These findings support a restorative, age-dependent hypothesis, but are not demonstrations of human infection prevention, general healthspan extension, or longer lifespan. (Kanemaru et al., 2026.)

Older experiments also support peripheral restoration: Zn-thymulin restored NK cytotoxicity in old-mouse cells in vitro, and 100 ng/day for five days in hydrocortisone-treated aged mice modestly augmented splenocyte responses (about 40% on average), without improving thymocyte responses. This supports effects outside the thymus and emphasizes biological context. (Muzzioli et al., 1992; Saha et al., 1995.)

Experimental Dose Range and Frequency

Representative animal experiments illustrate the breadth of exposure, rather than defining a human therapeutic range:

- Okamoto et al. (1993): 1 ng/kg daily for 3 days restored stress-impaired antibody production in mice toward normal. Doses of 1 ng–1 µg/kg did not stimulate antibody production in normal mice; the administration route is not specified in the abstract.
- Saha et al. (1995): 100 ng Zn-thymulin per aged, hydrocortisone-treated mouse, IP daily for 5 days, improved splenocyte responsiveness. This is a per-mouse dose, not 100 ng/kg.
- Henriques-Coelho et al. (2008): 100 ng/kg SC daily for 3 weeks improved inflammatory and cardiopulmonary endpoints in rats with monocrotaline-induced pulmonary hypertension; zinc-containing vehicle accounted for some effects.
- Novoselova et al. (2018): 1.5 mg/kg IP on days 1 and 5 modified an 8-day LPS-induced inflammatory process. This intermittent exception shows that dose frequency depends on the model; “septic inflammation” here is experimental endotoxemia, not a human infection trial.
- Kanemaru et al. (2026): 1.5 mg/kg IP daily, for one week in short immune experiments and up to four weeks in aged tumor-bearing mice, using peptide deliberately combined with equimolar zinc.

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In animal experiments, intraperitoneal thymulin at 0.1 or 10 mcg/kg increased circulating thymulin within 10 minutes; the increase remained apparent at 30 minutes, then declined rapidly, with levels below controls by 24 hours, supporting a transient circulating response model. (Buckingham et al., 1992.)

Daily exposure has stronger experimental precedent than large, widely spaced pulses.

Thymulin participates in thymocyte maturation and also acts on mature peripheral immune cells; conventional receptor-occupancy or receptor-desensitization model is not indicated.

Mechanistically, the wide range of experimental doses, quick clearing, and no receptor-desensitization model, taken together suggests frequency may impact efficacy as much as dose.

Dose & Patterns

There is no adequate human PK, exposure-response, or dose-ranging framework from which to optimize subcutaneous Thymulin dose and interval. (Amor et al., 1987.) Experimental doses in animal literature span a range factor of roughly 1.5 million. Using standard surface body scaling and then adjusting for mass, human doses of 0.1, 0.25, 0.5, and 1 mg all lie numerically inside that span, but this does not establish equal efficacy, safety, or a human therapeutic window. (Due to confusion with and Russian longevity protocols calling for Thymalin, a product unavailable on the gray market, anecdotal reports of subjects taking 5 - 10 mg doses of Thymulin for 10 or 20 days are common, with almost no adverse events reported.)

A few small, older Thymulin human studies exist, for example a RA rheumatoid-arthritis study evaluated thymulin at 1, 5, and 10 mg/day, but these small, condition-specific trials do not validate doses or indications.

A reasonable mechanistic extrapolated pattern is

- **Thymulin 0.25 mg daily for immune support / aging / healthspan**

These small, repeated doses may restore an age-depleted regulatory signal, and daily dosing has stronger experimental precedent. There is no evidence that 1 mg is superior to 0.25 mg for these aims. While there is no evidence that months or years of daily use are effective or safe, because thymulin is naturally occurring and declines with age, there is a mechanistic argument that continuous use is justified.

A different extrapolated pattern is

- **Thymulin 0.5 mg daily for up to 2 weeks, as an adjunct active infection therapy, beginning as early as possible after exposure or infection onset.**

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These exact doses, route, early-start timing, and duration are extrapolations, not clinically validated. The rationale is to support T/NK-cell function and immune regulation during a developing immune challenge, using frequent exposure consistent with experimental precedent.

Both proposed patterns mechanistically depend on biologically active Zn-thymulin.

Combining Thymulin and Thymosin Alpha-1

Thymulin and thymosin alpha-1 (Tα1) are potentially complementary rather than redundant. Tα1 has substantially stronger human clinical, dose, PK, and safety evidence and broadly modulates innate/adaptive immune coordination. Thymulin has weaker human evidence but a more direct relationship to age-related thymic endocrine decline, zinc biology, and inflammatory regulation.

A combined plausible hypothesis would pair Tα1's effects on antigen presentation and immune responsiveness with thymulin's potential restoration of deficient T/NK-cell function and restraint of excessive myeloid inflammation. Both peptides also engage regulatory pathways; their roles overlap and should not be divided into exclusive "stimulation" and "suppression" functions.

The combination of Tα1 and thymulin is mechanistically plausible, but without evidence.

Thymulin Benefits

- **Likely modulation of T-cell function and thymic signaling.** Zn-thymulin influences thymocyte maturation and peripheral T-cell markers in animal and in vitro studies.
- **Likely reduction of NF-κB-related inflammatory signaling in appropriate inflammatory contexts.** Zn-thymulin can modulate this pathway and reduce production of selected proinflammatory cytokines.
- **Plausible antiviral immune support**, based on T/NK-cell biology.
- **Plausible support for recovery of immune function after an immune stressor.** Zn-thymulin's effects on T-cell maturation and immune signaling provide a mechanistic rationale.
- **Plausible complementary effects with thymosin alpha-1.** Zn-thymulin and thymosin alpha-1 have partly distinct immunomodulatory actions, providing a mechanistic rationale for complementary effects; reported combined use is anecdotal.

Indications

- **Plausible potential application as adjunctive immune support for recurrent infections.** Zn-thymulin's role in T-cell maturation provides a mechanistic rationale; reported use is anecdotal.
- **Plausible potential application as adjunctive support during postviral recovery.** Zn-thymulin's immunomodulatory effects provide a mechanistic rationale; reported use is anecdotal.

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- **Plausible support for age-related immune dysfunction.** Declining endogenous active thymulin, older-animal immune-restoration studies, and experiments in aged human PBMCs support investigation; they do not establish clinical healthspan benefit.
- **Plausable reduction of age-associated myeloid inflammatory signaling**. This has been demonstrated in aged human cells ex vivo and aged mice. (Kanemaru et al., 2026.)
- **Plausible adjunctive use with thymosin alpha-1 for complementary immunomodulation.** Their partly distinct immunomodulatory actions provide a mechanistic rationale; reported combined use is anecdotal.
- **Plausible potential adjunct in selected chronic inflammatory or autoimmune contexts.**

Cautions

- **Active or recent malignancy:** use should be reviewed with the treating oncologist. Improved tumor control and checkpoint-treatment response in aged mice do not establish safety or benefit in human cancer or justify modifying cancer treatment.
- **Unstable or uncontrolled autoimmune disease with active severe flares:** avoid unsupervised immune-active interventions.
- **Unsupervised use in people on potent immunosuppressants or biologics** (risk of unpredictable interactions).
- **Pregnancy and breastfeeding** — insufficient safety data.
- **Uncorrected zinc deficiency** may impair active thymulin biology. Adequate zinc status does not establish that an injected zinc-free product is equivalent to preformed Zn-thymulin.

Repeated Exposure, Diminished Response, and Evidence Limits

Repeated animal administration, including daily schedules lasting up to approximately 4 weeks, has produced biological responses without obvious loss of effect offers limited reassurance against rapid tachyphylaxis.

There are no adequate months- or years-long human data for the proposed chronic regimen. Human pathway adaptation, immunogenicity, safety, and sustained benefit remain unknown. Daily dosing has strong experimental precedents, without establishing an optimal long-term schedule.

Adequate Zinc Status and Thymulin

Zinc deficiency can reduce endogenous active thymulin even when zinc-free peptide remains present. Correcting inadequate intake or deficiency can restore some activity; this is distinct from proving activation of an injected product. Zinc supplementation should address nutritional needs, not be assumed to guarantee quantitative conversion of zinc-free FTS to Zn-thymulin. (Prasad et al., 1988.)

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Zinc intake should generally meet established dietary requirements, which are 8 mg daily for adult women and 11 mg daily for adult men. The adult tolerable upper intake level of 40 mg daily includes zinc from food, supplements, and medications.

Serum zinc, serum copper, and ceruloplasmin may help assess zinc-copper balance, interpreted in clinical context. Serum zinc is influenced by inflammation and other factors. Excess zinc can impair copper absorption and immune function; increasing zinc beyond nutritional needs is not an established way to improve thymulin effects. (NIH Office of Dietary Supplements.)

Personal Observation: Morning Congestion

I experienced chronic low-grade postnasal drip and morning congestion for as long as I can remember. Since beginning Thymulin, those symptoms resolved. This is an uncontrolled personal observation; it does not establish causation or validate an intermittent dose schedule.

A speculative explanation is **reduced mucosal inflammatory signaling**. In vitro Zn-thymulin reduces NF-κB activation and LPS-induced IL-1β release in alveolar epithelial cells; thymulin gene therapy also reduced airway inflammation and mucus-related pathology in an allergic-asthma mouse model. Extending these findings to nasal congestion or zinc-free SC peptide is uncertain, and conversion to Zn-thymulin was not measured in this observation. (Haddad, 2009; da Silva et al., 2014.)

Why Zn-thymulin Might Help with Serious Infection

Zn-Thymulin has a plausible rationale in severe infection because it is involved in T-cell maturation and function. It supports Th1 immunity, including IL-2 production, which in turn supports cytotoxic T cells and NK cells needed to eliminate infected cells. In **severe infection accompanied by immune exhaustion, restoring these functions could theoretically improve pathogen clearance**. Zn-thymulin also has anti-inflammatory actions, including suppression of NF-κB, p38 MAPK and pro-inflammatory cytokines, so it could potentially strengthen effective cellular immunity without simply increasing systemic inflammation. **The rationale may be stronger in older or zinc-deficient patients**, because aging and zinc deficiency are associated with lower Zn-thymulin and impaired cellular immunity.

Experimental LPS studies support anti-inflammatory activity, but do not show improved human pathogen clearance or survival in sepsis. Severe infection requires established medical care; the acute research pattern above has not been validated for serious infection. (Novoselova et al., 2018.)

Why Zn-thymulin Might Help with Minor Infection

For a minor viral infection, Zn-thymulin could theoretically help by strengthening cellular antiviral immunity. It promotes T-cell differentiation and function and is associated with Th1 signaling, including IL-2 and IFN-γ; these pathways support NK cells and cytotoxic T cells that recognize

and destroy virus-infected cells. **This rationale is probably weakest in a healthy young person whose thymus and cellular immune response are already functioning well, but becomes more plausable in older people, because of thymic involution, declining Zn-thymulin activity, reduced NK-cell activity and weaker cell-mediated immunity accompany aging.**

The proposed 0.5-mg daily short course and early start are merely mechanistic choices.

Why Zn-thymulin Might Help with Autoimmune Diseases

Zn-thymulin may also make biological sense in autoimmune disease because its function is immunoregulatory rather than simply immunostimulatory. It influences differentiation and activity of multiple T-cell populations, with particularly **notable effects historically reported on suppressor/regulatory T-cell function, which helps prevent immune responses against the body's own tissues**. Zn-thymulin also has anti-inflammatory effects, including inhibition of NF-κB and p38 MAPK signaling and reductions in pro-inflammatory cytokine activity. The plausable result is therefore **better immune regulation: strengthening deficient T-cell functions where necessary while restraining inappropriate inflammatory or autoreactive responses**. Altered or reduced Zn-thymulin activity has been reported in autoimmune diseases, but evidence that administering thymulin improves human autoimmune disease is sparse, making the therapeutic argument substantially more speculative than the underlying biology.

Lupus-prone mouse studies reported reductions in anti-DNA antibodies and renal pathology, supporting a plausible immunoregulatory rationale. No quantitative autoimmune injection schedule is established here; these findings do not validate a human autoimmune regimen. (Israel-Biet et al., 1983; Lunin et al., 2015.)

Hashimoto's thyroiditis

Zn-thymulin's effects on T-cell maturation and NF-κB-related inflammatory signaling provide a plausible mechanistic rationale for modulating autoimmune inflammation in Hashimoto's thyroiditis. Anecdotal users report reduced inflammatory symptoms and improved resilience.

Rheumatoid arthritis (RA)

Small, older human studies provide preliminary evidence in rheumatoid arthritis. Zn-thymulin improved abnormal T-cell subset markers in vitro in eight of nine patients, and an open clinical report described improvement. Two randomized, double-blind, placebo-controlled trials evaluated thymulin at 1, 5, and 10 mg/day; 5 mg/day produced the best reported clinical response with minimal adverse effects. These findings suggest possible symptomatic benefits, but do not establish modern routine RA treatment or validate the proposed healthspan or infection patterns. (Faure et al., 1984, 1987; Amor et al., 1987.)

Some users report reduced morning stiffness or milder flares, but these reports are anecdotal.

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Rheumatoid arthritis with interstitial lung disease (RA-ILD)

Zn-thymulin's immunomodulatory and NF-κB-related effects provide a plausible rationale for modulating the inflammatory component of RA-ILD. Limiting fibrosis during inflammatory lung injury is also plausible, supported by reduced fibrosis with early FTS treatment in an experimental lung-injury model. This does not establish reversal of existing lung scarring.

Biological Mechanism

Thymic epithelial cells produce FTS. Binding zinc forms biologically active Zn-thymulin, an endocrine and paracrine regulator with both intrathymic and extrathymic actions. Exogenous effects on peripheral lymphocytes and myeloid cells do not depend on thymic regrowth.

Mechanistically, Zn-thymulin exerts multiple effects:

- **Modulation of thymocyte maturation and peripheral T-cell phenotype.** Zn-thymulin influences thymocyte maturation and can alter peripheral T-cell markers. Effects on thymic selection, the peripheral T-cell repertoire, and central tolerance are plausible, but the specific receptor and signaling pathways remain incompletely defined.
- **Modulation of NF-κB-related inflammatory signaling.** Zn-thymulin likely reduces NF-κB-related inflammatory signaling in appropriate inflammatory contexts. In vitro research shows reduced NF-κB nuclear translocation and lipopolysaccharide-induced IL-1β release in alveolar epithelial cells, while animal studies support modulation of this pathway during inflammation. Broader anti-inflammatory effects are likely in appropriate inflammatory contexts, including reductions in elevated circulating proinflammatory cytokines.
- **Historical suppressor-cell findings, T-cell differentiation, and modulation of inflammatory signaling support immune regulation rather than uniform stimulation.** Older suppressor-cell assays should not automatically be equated with a defined modern regulatory T-cell phenotype. cAMP-dependent and other downstream pathways have been described in experimental systems, without a complete unified receptor model.
- **Endogenous thymulin participates in bidirectional communication between the thymus, immune system, and neuroendocrine system.** Experimental effects involve pituitary signaling and vary with age and context. (Buckingham et al., 1992; Reggiani et al., 2009.)

These effects provide a rationale for investigating potential applications in recurrent infections, immune dysfunction, and selected inflammatory or autoimmune conditions.

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